

2 hours on examples

Problem 2 (AIME 2009/15)

By the incenter-excenter lemma, the circumcenter of $BI_B D$ is O_B the intersection of the angle bisector of $\angle BAD$ and (ABD) . The circumcenter $BI_C D$ is O_C the intersection of the angle bisector of $\angle CAD$ and (ACD) .

By radical axis, $O_B O_C$ is the perpendicular bisector of DP . First note that,

$$\angle O_B P O_C = \angle O_B P D + \angle D P O_C = 90 - \frac{1}{2} \angle D O_B P + 90 - \frac{1}{2} \angle D O_C P = 180 - \angle C B P - \angle B C P = \angle B P C$$

Therefore,

$$\begin{aligned} \angle B A C &= \angle B A D + \angle C A D = (180 - \angle D O_B B) + (180 - \angle D O_C C) = 2(\angle B D O_B + \angle C D O_C) \\ &= 360 - 2\angle O_C D O_B = 360 - 2\angle O_B P O_C = 360 - 2\angle B P C \\ &\Leftrightarrow \angle B P C = 180 - \frac{1}{2} \angle B A C \end{aligned}$$

Therefore, P lies on a circle with chord BC . Obviously, the area of BPC is maximized when P is the midpoint of minor arc BC . At that point, $\angle C B P = \angle B C P = \frac{1}{4} \angle B A C$.

We may calculate,

$$\cos(\angle B A C) = \frac{16^2 + 10^2 - 14^2}{2 \cdot 16 \cdot 10} = 0.5 \Leftrightarrow \angle B A C = 60^\circ$$

$$\text{We get, } [BPC] \leq \frac{1}{2} \cdot 14 \cdot 7 \cdot \tan\left(\frac{1}{4} \angle B A C\right) = 49 \tan(15^\circ) = 49 \frac{\sin(30)}{1 + \cos(30)} = 49 \frac{1}{2 + \sqrt{3}} = 98 - 49\sqrt{3}.$$

Problem 4 (PRIMES 2019 entrance exam)

The average area is $\frac{1}{3}$ the total area. The total area is obviously maximized when X is near an endpoint, so $S < \frac{1}{3}Y$ where Y is the area of ABC .

We claim the lower bound is $\frac{1}{9}Y$, which can obviously be achieved when X is the centroid. Let a be the ratio of the length of the altitude from X to BC to that of A to BC . Similarly define b, c . We see that,

$$a = \frac{[XBC]}{[ABC]}; b = \frac{[AXC]}{[ABC]}; c = \frac{[ABX]}{[ABC]}$$

From this, we get that $a + b + c = 1$. The average area of triangles T_a, T_b, T_c is simply,

$$Y \cdot \frac{a^2 + b^2 + c^2}{3} \leq Y \cdot \frac{1}{9}$$

By Power-Mean, this is minimized when $a = b = c = \frac{1}{3}$, so we're done.

The sum of the lengths of the segments parallel to BC is the length of BC , and same can be said for the other 2. Therefore, the average perimeter p is always $\frac{1}{3}$ that of triangle ABC .

6 clubs; 4 hours

Problem 7 (NIMO 5.8)

We let $y := BZ \Leftrightarrow BX = y + 9$; $x := AX \Leftrightarrow BX = x + 6$. By Pythag,

$$AB^2 = 5^2 + 12^2 = (x + 6)^2 + y^2 = x^2 + (y + 9)^2$$

We'd have,

$$x^2 + y^2 + 18y = 88;$$

$$x^2 + y^2 + 12x = 133$$

Subtracting, we'd get,

$$12x - 18y = 45 \Leftrightarrow x = \frac{3}{2}y + \frac{15}{4}$$

Therefore,

$$x^2 + y^2 + 18y = 88 \Leftrightarrow \frac{13}{4}y^2 + \frac{117}{4}y + \frac{225}{16} = 88$$

$$\Leftrightarrow \frac{13}{4}y^2 + \frac{117}{4}y - \frac{1183}{16} = 0$$

$$\Leftrightarrow 4y^2 + 36y - 91 = 0$$

$$\Leftrightarrow y = \frac{-36 \pm \sqrt{36^2 + 4 \cdot 4 \cdot 91}}{8} = \frac{-36 \pm 4\sqrt{81 + 91}}{8}$$

Taking the positive result,

$$y = \frac{-9 + 2\sqrt{43}}{2} \Leftrightarrow y + 9 = \frac{9 + 2\sqrt{43}}{2}$$

$$\text{We then get, } x = \frac{3}{2}y + \frac{15}{4} = \frac{-27 + 6\sqrt{43}}{4} + \frac{15}{4} = \frac{-6 + 3\sqrt{43}}{2} \Leftrightarrow x + 6 = \frac{6 + 3\sqrt{43}}{2}.$$

By Ptolemy's,

$$13 \cdot YZ + 12 \cdot y = 5 \cdot (x + 6)$$

$$\Leftrightarrow YZ = \frac{5 \left(\frac{6 + 3\sqrt{43}}{2} \right) - 12 \left(\frac{-9 + 2\sqrt{43}}{2} \right)}{13} = \frac{138 - 9\sqrt{43}}{26}$$

We may also get,

$$13 \cdot XY + 5 \cdot x = 12 \cdot (y + 9)$$

$$\Leftrightarrow XY = \frac{12\left(\frac{9+2\sqrt{43}}{2}\right) - 5\left(\frac{-6+3\sqrt{43}}{2}\right)}{13} = \frac{138+9\sqrt{43}}{26}$$

Therefore,

$$OX + OZ + YZ + XY = 13 + \frac{138}{13} = \frac{307}{13}$$

Problem 9 (PUMaC 2016 G7)

Note that,

$$\angle ABD = 180 - \angle DYA = \angle EYD;$$

$$\angle CAB = \angle CDB = \angle DEA$$

Therefore, DEY and XAB are directly similar.

By dropping an altitude from A to BC , we see that ABX consists of an 8-15-17 and 6-8-10 and hence has area 84. Note that,

$$AB = DY = 10$$

Therefore, DEY 's area is $\left(\frac{10^2}{21^2}\right)$ that of 84, which evaluates to $\frac{400}{21}$.

Problem 11 (USAMO 2020/1)

Note that BX is the radical axis of the circumcircle and (BDX) , DX is that of (BDX) , (ADX) , and AX is that of (ADX) and the circumcircle. Therefore, OO_2 is the perpendicular bisector of BX , OO_1 is that of AX , and O_1O_2 is that of DX .

Claim: OO_1O_2 is oppositely similar to CBA .

We may get $\angle O_1O_2O = \angle CXB = \angle A$; $\angle O_2O_1O = \angle AXC = \angle B$.

Claim: O_1XO_2 is directly similar to AXB .

We may get,

$$\angle XO_2O_1 = \frac{1}{2}\angle XO_2D = \angle XBA; \angle XO_1O_2 = \frac{1}{2}\angle XO_1D = \angle BAX$$

We can let C' be the reflection of C over AB such that we get OO_1O_2 is directly similar to CBA . Note that $[C'BA] = [CBA]$, so we minimize the area of $C'BA$ to get the answer.

Since O_1XO_2O is a directly similar quadrilateral to AXO_2C' , we are effectively minimizing, the ratio r of the edges. However, we see that,

$$r = \frac{OO_2}{C'B} = \frac{O_1X}{AX}$$

Since AO_1X is an isosceles triangle, we find that $O_1X \geq \frac{AX}{2}$. Therefore, $r \geq \frac{1}{2}$ and the area of OO_1O_2 is at least $\frac{1}{4}$ that of ABC .

The equality case occurs when AO_1X are collinear,

$$\angle O_1XA = 0 \Leftrightarrow \angle ADX = 90^\circ$$

Therefore, the equality case, the minimum is attained when D is the foot of the perpendicular from C to AB .

15 clubs; 8 hours

Problem 14 (AIME II 2021/14)

We let $\angle BCA = 2x$; $\angle ABC = 13x$; $\angle YOX = 17x$. We let M be the midpoint of BC and have BC intersect the tangent at A at point P outside the triangle.

$$\begin{aligned} 17x = \angle YOX &= \angle GOX + \angle YOG = 180^\circ - \angle GAX + \angle YMG = \angle PAM + \angle PMA \\ &= \angle BAP + \angle ABP = 180^\circ - 11x \end{aligned}$$

Therefore, $28x = 180^\circ \Leftrightarrow x = \frac{45}{7}^\circ$.

$$\angle A = 180 - 2x - 13x = 13x = \frac{585}{7}^\circ$$

Problem 16 (JMO 2016/5)

We use ghost points, let $AO \cap PQ = O'$. We prove $O' = O$.

$$\text{Claim: } \sin(B) \cdot \sin(C) = \frac{\sqrt{2}}{2}$$

We may get,

$$\begin{aligned} \frac{\sqrt{2}}{2} &= \frac{AO}{AH} \\ \Leftrightarrow \frac{\sqrt{2}}{2} &= \frac{AH}{2R} \end{aligned}$$

Using Law of Sines,

$$\Leftrightarrow \frac{\sqrt{2}}{2} = \frac{2R \cdot \sin(B) \cdot \sin(C)}{2R} = \sin(B) \cdot \sin(C)$$

Claim: $AO' \perp PQ \Leftrightarrow AO \perp PQ$

By cyclic quads, we can get,

$$\angle ABC = \angle AHP = \angle AQP = \angle AQO'$$

However, $\angle OAC = \angle O'AC = 90 - \angle B$. Therefore, $\angle AO'Q = 90^\circ$.

Therefore,

$$\frac{AO'}{AH} = \frac{AP \sin(C)}{AH} = \sin(B) \sin(C) = \frac{\sqrt{2}}{2} = \frac{AO}{AH}$$

Since P and Q are both inside ABC , O' must also be inside ABC and is hence on the same side of A as O . We therefore get $O' = O$, meaning P, O, Q are collinear.

25 clubs; 11 hours

Problem 17 (AIME 2010/15)

We let, $x := BM$; $15 - x = CM$. Using Stewart's,

$$\begin{aligned} ax(15 - x) + ad^2 &= b^2x + c^2(15 - x) \\ \Leftrightarrow 15(x(15 - x) + d^2) &= 169x + 144(15 - x) \\ \Leftrightarrow d^2 &= \frac{3x^2 - 40x + 432}{3} \end{aligned}$$

Using $A = sr$,

$$\begin{aligned} \frac{[ABM]}{[ACM]} &= \frac{s_{ABM}}{s_{ACM}} \\ \Leftrightarrow \frac{x}{15 - x} &= \frac{x + d + 12}{15 - x + d + 13} \\ \Leftrightarrow x(15 - x) + xd + 13x &= x(15 - x) + (15 - x)d + 12(15 - x) \\ \Leftrightarrow xd + 13x &= 15d - xd + 180 - 12x \\ \Leftrightarrow d &= \frac{-25x + 180}{2x - 15} \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{3x^2 - 40x + 432}{3} &= \frac{625x^2 - 9000x + 32400}{4x^2 - 60x + 225} \\ \Leftrightarrow 12x^4 - 340x^3 + 4803x^2 - 34920x + 97200 &= 1875x^2 - 27000x + 97200 \end{aligned}$$

If $x = 0$, then d is negative, so $x \neq 0$.

$$\Rightarrow 3x^3 - 85x^2 + 732x - 1980 = 0$$

Plugging in x , we see that $x = 6$ works, we may then factor the entire thing out.

$$(x - 6)(x - 15)(3x - 22) = 0$$

However, at $x = 15$, $x = 6$, we'd have $d = \frac{-25x+180}{2x-15}$ is negative, so there'd be an extraneous solution.

At $x = \frac{22}{3}$, we'd have $d = \frac{-25 \cdot \frac{22}{3} + 180}{2 \cdot \frac{22}{3} - 15} = 10$. This works out. We'd also get,

$$CM = 15 - \frac{22}{3} = \frac{23}{3}$$

Therefore, $\frac{BM}{CM} = \frac{22}{23}$.

Problem 22 (CMIMC 2023 G8) (100% Pure Complex)

We use complex coordinates. Let, ABC be on the unit circle with AB as the real axis. Let the coordinates of A be -1 , B be 1 , O be 0 . We let $C = x^2$. Of the two choices, we take the one where $M = ix$; $N = x$.

$$\begin{aligned} p &= \frac{(1)(x^2)(-1 + ix) - (-1)(ix)(x^2 + 1)}{(1)(x^2) - (-1)(ix)} = \frac{-x^2 + ix^3 + ix^3 + ix}{x^2 + ix} \\ &= \frac{2ix^2 + i - x}{x + i} = \frac{(2ix + 1)(x + i)}{x + i} \end{aligned}$$

Since $x \neq -i$ for the triangle to not be degenerate,

$$p = 2ix + 1$$

To avoid confusion, let y be the complex coordinates of I .

$$\begin{aligned} y &= \frac{(1)(ix)(x - 1) - (-1)(x)(1 + ix)}{(1)(ix) - (-1)(x)} = \frac{ix^2 - ix + x + ix^2}{ix + x} \\ &= \frac{2ix - i + 1}{i + 1} = \frac{(2ix - i + 1)(1 - i)}{2} = \frac{2ix - i + 1 + 2x - 1 - i}{2} \\ &= xi + x - i \end{aligned}$$

We want O, I, P to be collinear, so we write,

$$\begin{aligned} \frac{p}{y} &= \frac{\bar{p}}{\bar{y}} \\ \Leftrightarrow \frac{2ix + 1}{xi + x - i} &= \frac{\overline{2ix + 1}}{\overline{xi + x - i}} \\ \Leftrightarrow \frac{2ix + 1}{xi + x - i} &= \frac{-\frac{2i}{x} + 1}{-\frac{i}{x} + \frac{1}{x} + i} \end{aligned}$$

$$\Leftrightarrow \frac{2ix + 1}{xi + x - i} = \frac{x - 2i}{ix - i + 1}$$

$$\Leftrightarrow -2x^2 + (2 + 3i)x + (1 - i) = x^2(1 + i) + (2 - 3i)x - 2$$

$$\Leftrightarrow 0 = (3 + i)x^2 + (-6i)x + (-3 + i)$$

Solving, we get $x = i, \frac{3}{5} + \frac{4i}{5}$. The first solution is extraneous since that'd mean $a = c$, impossible. For the second solution, we'd get,

$$c = \left(\frac{3}{5} + \frac{4i}{5}\right)^2 = -\frac{7}{25} + \frac{24}{25}i$$

Therefore, the height of C to AB is $\frac{24}{25}$. The area is hence,

$$[ABC] = \frac{24}{25}$$

35 clubs; 13.5 hours

Problem 23 (22BISSUE15)

Claim: The angle bisectors of B and C intersect at P , and P is on AD .

We let $\angle A = x$; $\angle B = 2x$; $\angle C = 2y$; $\angle D = y$. Let P be the intersections of the angle bisectors of B and C . We show that A, P, D are collinear.

$$\angle APB = 180 - 2x; \angle BPC = 180 - x - y; \angle CPA = 180 - 2y$$

The sum of the three angles is $540 - 3x - 3y = 540 - \angle A - \angle B - \angle C - \angle D = 180$, so we're done.

With the isosceles triangles, we let $a = AP = BP$; $b = DP = CP$. We may get the following formulas using $y = 120 - x$.

$$2a \cos(x) = 3;$$

$$2b \cos(120 - x) = 5;$$

$$\frac{a}{\sin(120 - x)} = \frac{b}{\sin(x)}$$

We let E be the base of the perpendicular from P to E . Note that by congruent triangles, $BE = 1.5$; $CE = 2.5$. Therefore, the area of $ABCD$ is thrice that of BPC .

$$PE = a \sin(x)$$

$$\Leftrightarrow PE = \frac{3}{2} \tan(x)$$

$$\Leftrightarrow [ABCD] = 9 \tan(x)$$

Using the third equation,

$$\begin{aligned} &\Leftrightarrow \frac{3}{2 \cos(x) \sin(120 - x)} = \frac{5}{2 \sin(x) \cos(120 - x)} \\ &\Leftrightarrow \frac{3}{\cos(x) \cos(x) \sqrt{3} + \cos(x) \sin(x)} = \frac{5}{-\sin(x) \cos(x) + \sin(x) \sin(x) \sqrt{3}} \\ &\Leftrightarrow -3 \sin(x) \cos(x) + 3\sqrt{3} \sin^2(x) = 5\sqrt{3} \cos^2(x) + 5 \sin(x) \cos(x) \end{aligned}$$

Note that $x, y < 90$ since otherwise we'd have an angle greater than 180° , so we let,

$$X = \sin(x) \Rightarrow \sqrt{1 - X^2} = \cos(x)$$

Therefore, the previous equation,

$$\begin{aligned} &\Leftrightarrow (3X^2 - 5(1 - X^2))\sqrt{3} = 8X\sqrt{1 - X^2} \\ &\Leftrightarrow 8X^2\sqrt{3} - 5\sqrt{3} = 8X\sqrt{1 - X^2} \\ &\Leftrightarrow 192X^4 - 240X^2 + 75 = 64X^2 - 64X^4 \\ &\Leftrightarrow 256X^4 - 304X^2 + 75 = 0 \\ &\Leftrightarrow \sin(x)^2 = \frac{304 + \sqrt{304^2 - 4 \cdot 75 \cdot 256}}{512} = \frac{19 + \sqrt{19^2 - 300}}{32} = \frac{19 + \sqrt{61}}{32} \\ &\Leftrightarrow \cos^2(x) = \frac{13 - \sqrt{61}}{32} \\ &\Rightarrow \tan^2(x) = \frac{19 + \sqrt{61}}{13 - \sqrt{61}} = \frac{308 + 32\sqrt{61}}{108} = \frac{77 + 8\sqrt{61}}{27} \\ &\Leftrightarrow \left(\frac{9 \tan(x)}{\sqrt{3}} \right)^2 = 77 + 8\sqrt{61} \end{aligned}$$

Noting that $\tan(x)$ is positive,

$$\begin{aligned} &\Leftrightarrow \frac{9 \tan(x)}{\sqrt{3}} = 4 + \sqrt{61} \\ &\Leftrightarrow [ABCD] = 4\sqrt{3} + \sqrt{183} \end{aligned}$$

44 clubs; 15.5 hours